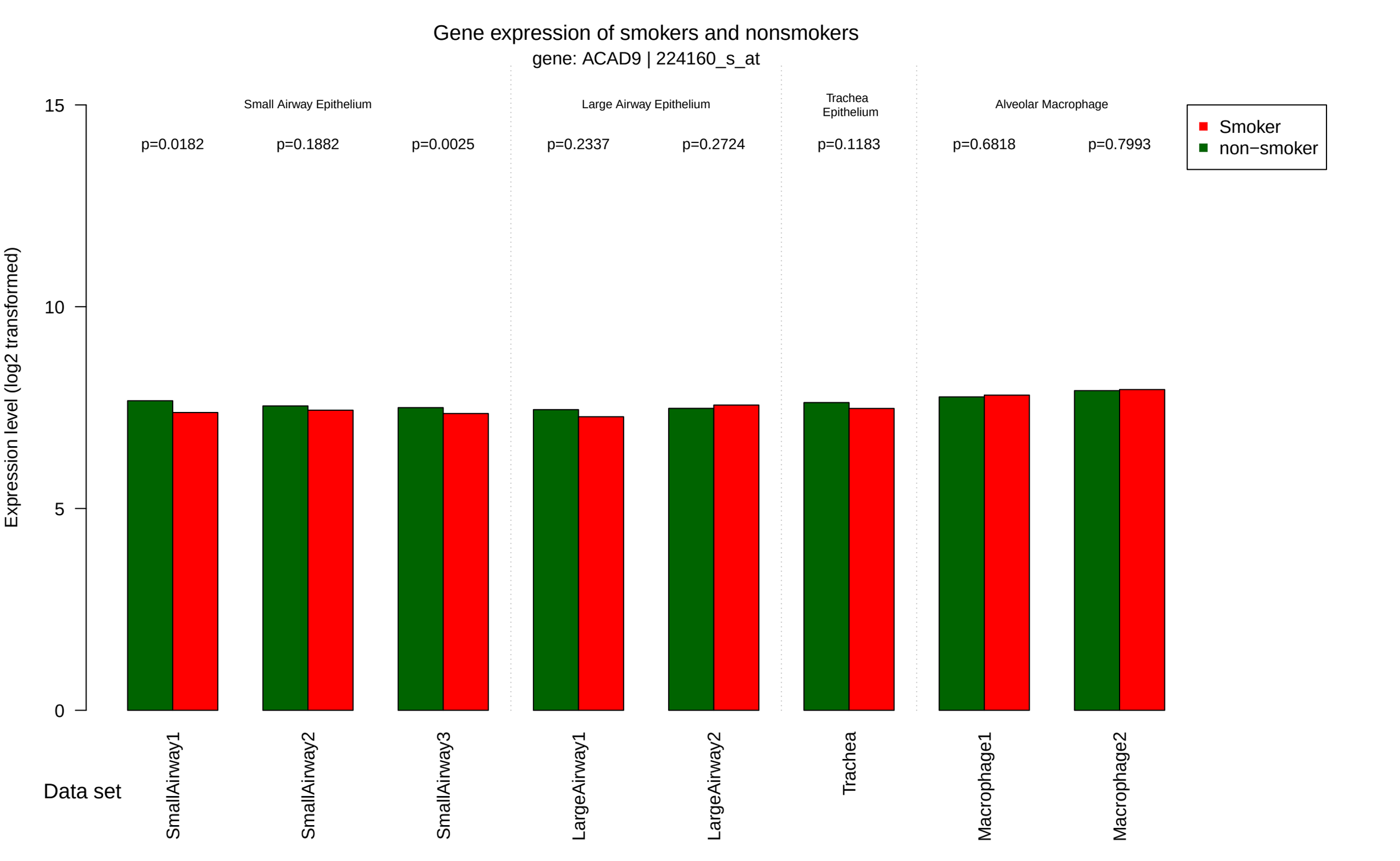
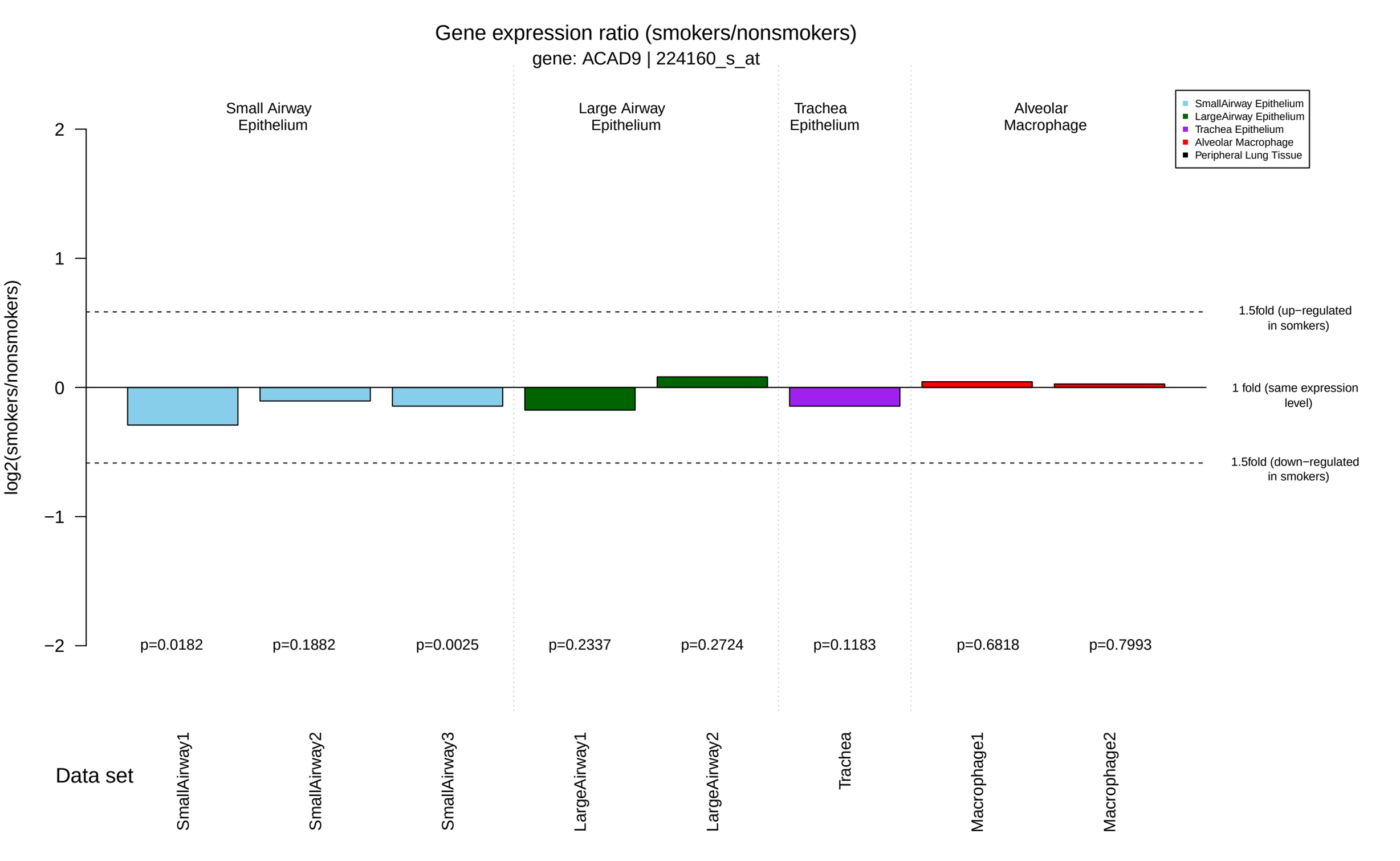
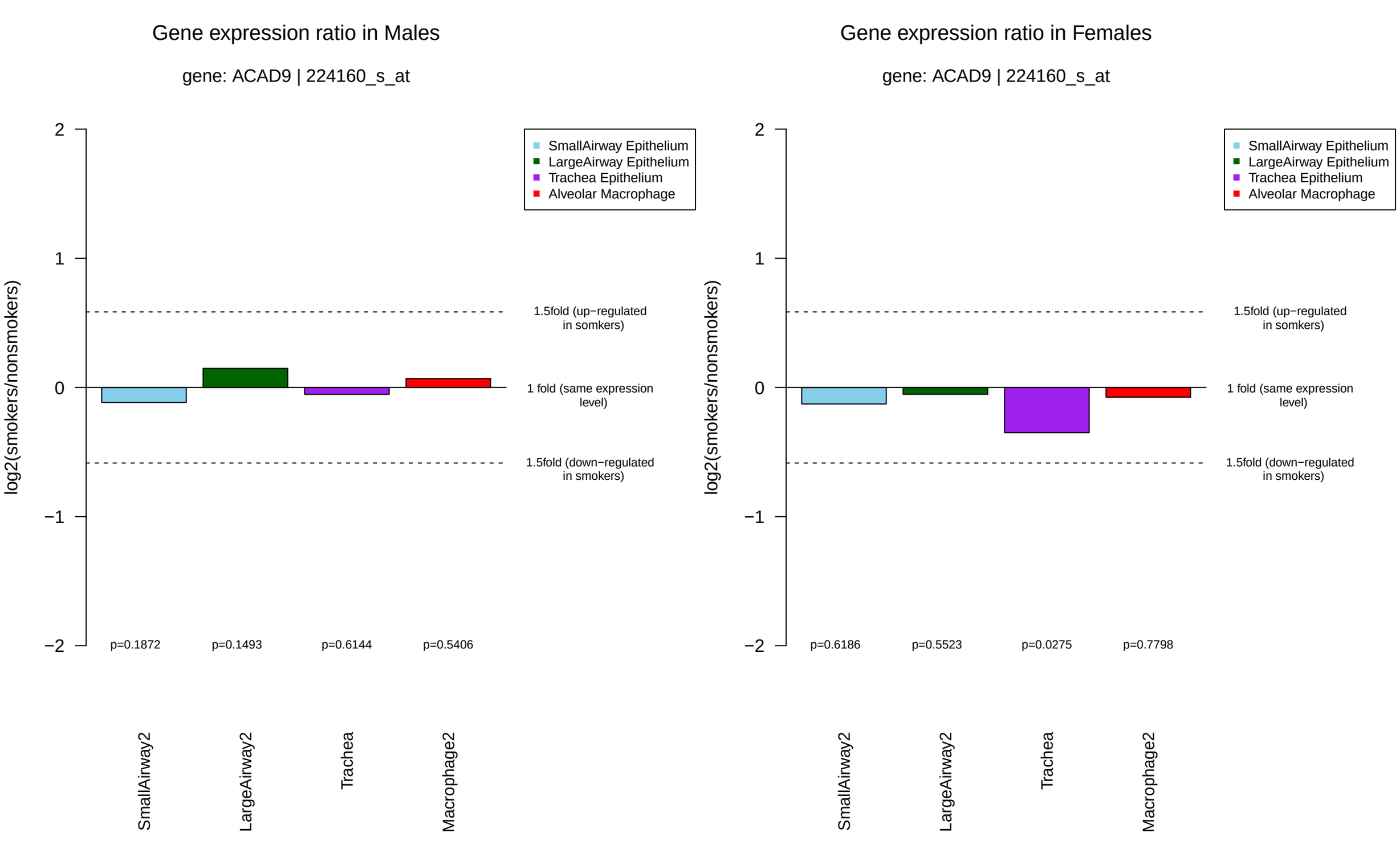
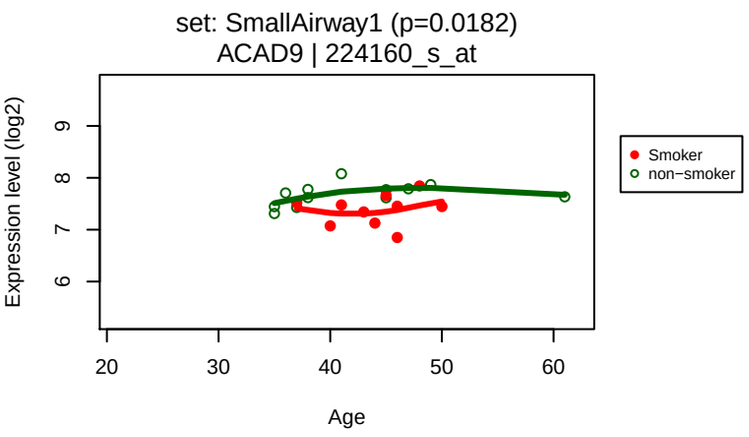
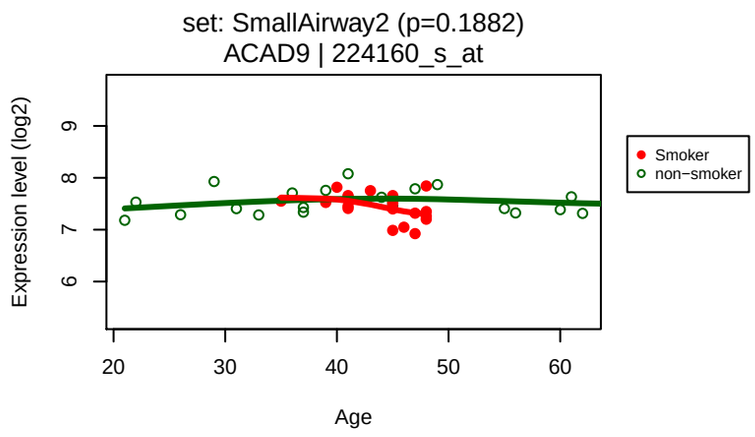
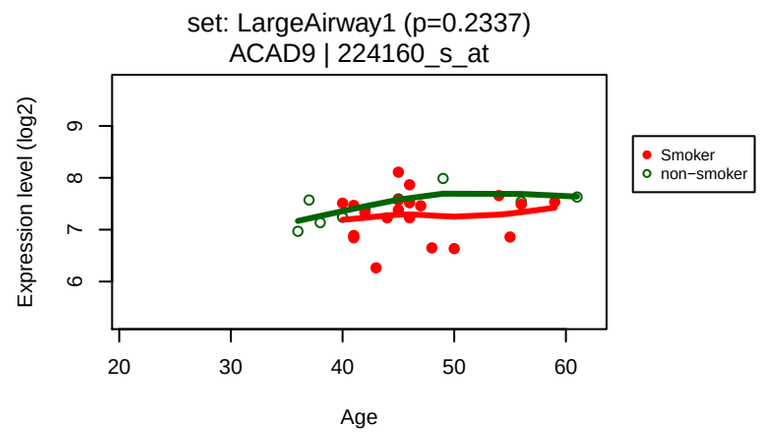

Gene expression of smokers and nonsmokers
gene: ACAD9 | 224160_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0182
p=0.1882
p=0.0025
p=0.2337
p=0.2724
p=0.1183
p=0.6818
p=0.7993
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

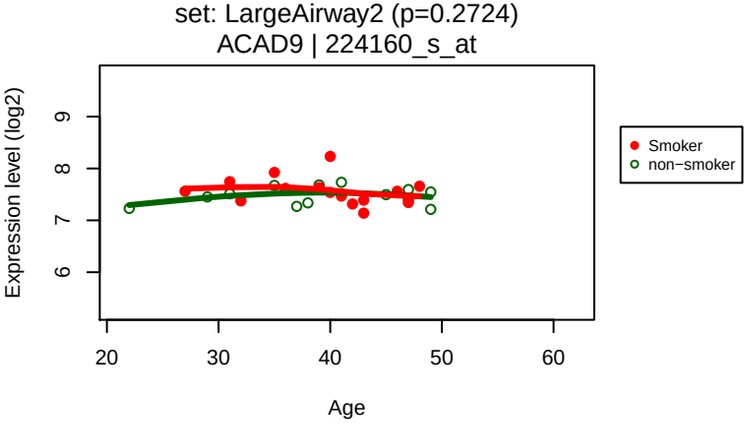

Gene expression ratio (smokers/nonsmokers)
gene: ACAD9 | 224160_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0182
p=0.1882
p=0.0025
p=0.2337
p=0.2724
p=0.1183
p=0.6818
p=0.7993
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

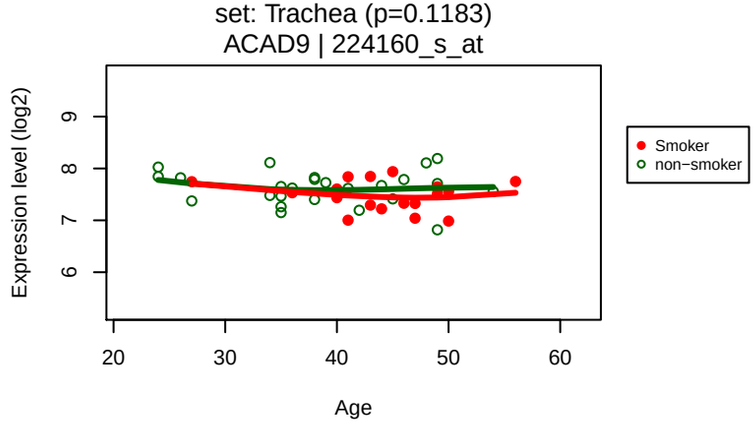

Gene expression ratio in Males
gene: ACAD9 | 224160_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1872
p=0.1493
p=0.6144
p=0.5406
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAD9 | 224160_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6186
p=0.5523
p=0.0275
p=0.7798
SmallAirway2
LargeAirway2
Trachea
Macrophage2

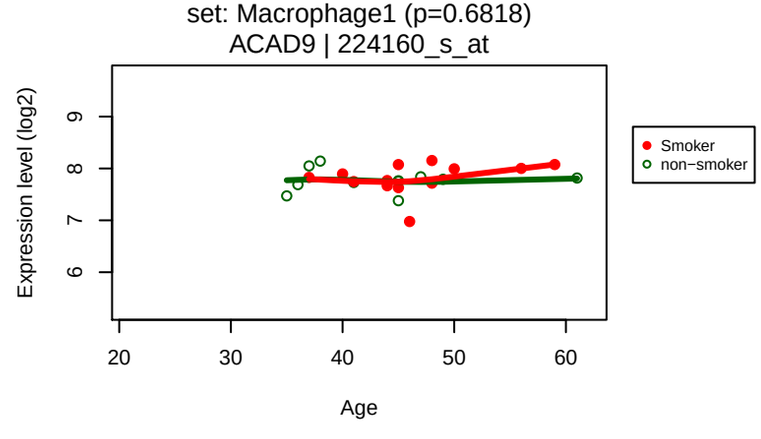

set: SmallAirway1 (p=0.0182)
ACAD9 | 224160_s_at
Expression level (log2)
Age
Smoker
non−smoker

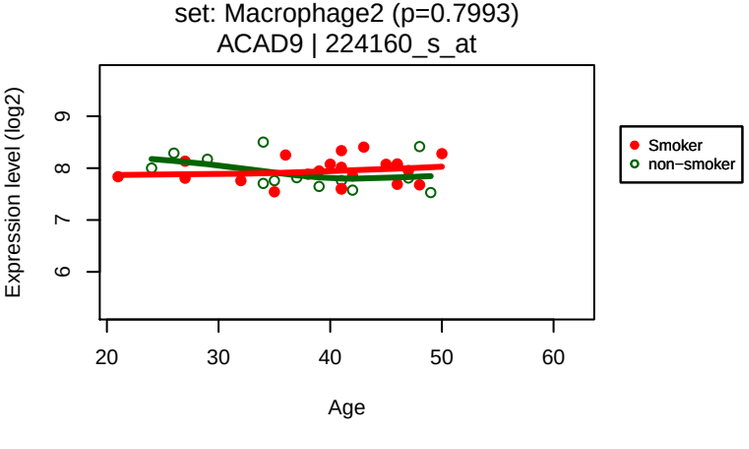

set: SmallAirway2 (p=0.1882)
ACAD9 | 224160_s_at
Expression level (log2)
Age
Smoker
non−smoker

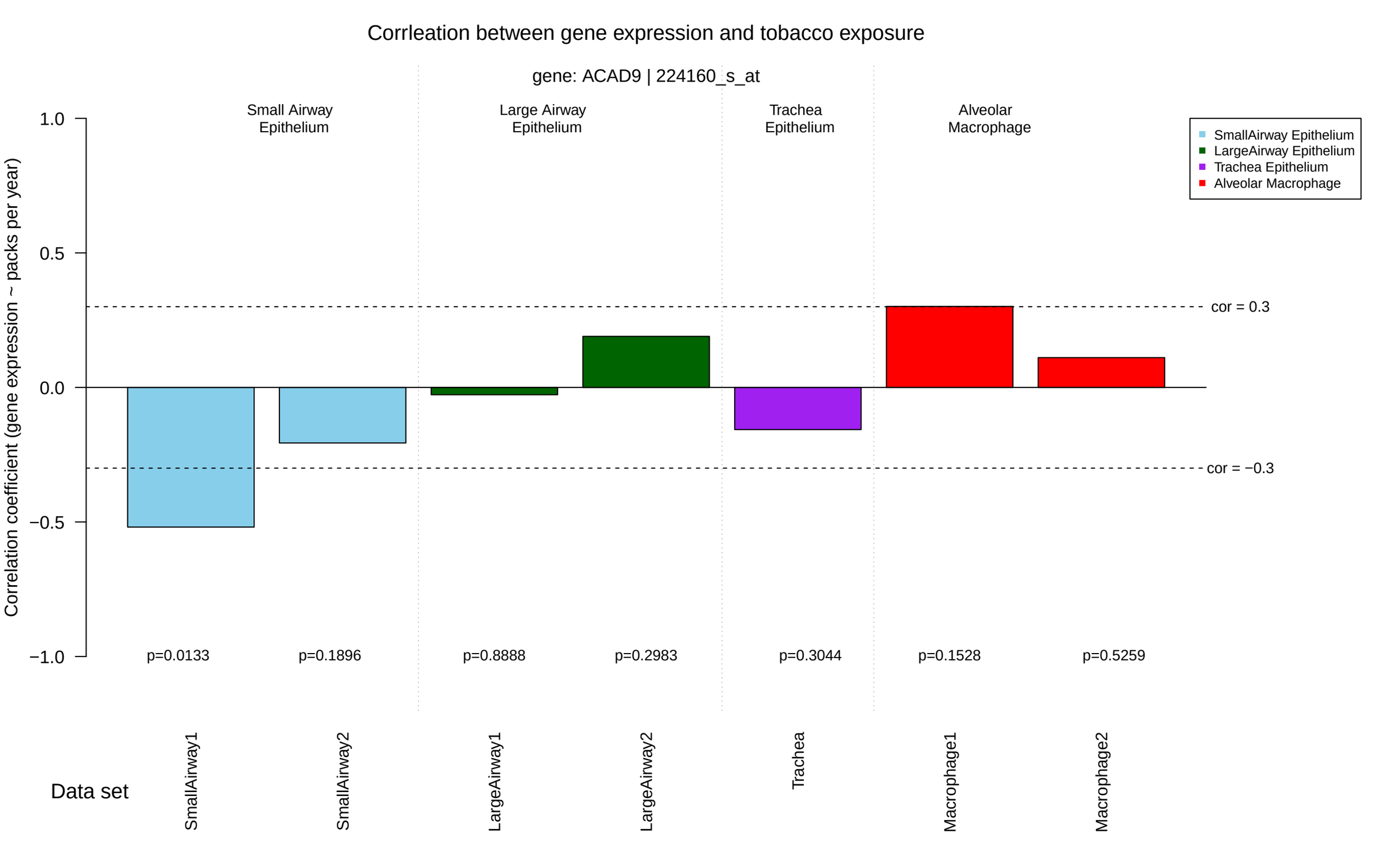

set: LargeAirway1 (p=0.2337)
ACAD9 | 224160_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2724)
ACAD9 | 224160_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1183)
ACAD9 | 224160_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6818)
ACAD9 | 224160_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7993)
ACAD9 | 224160_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAD9 | 224160_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0133
p=0.1896
p=0.8888
p=0.2983
p=0.3044
p=0.1528
p=0.5259
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2